nba_api calls: what to use and what to drop

Audit of the 145 calls in the `nbaapi analysis/data understanding/` notebooks. Tested **2026-10-25** with **nba_api 1.12.0** on the **2025-26** season.

At a glance

Every call was re-run with the notebooks' own inputs (Wembanyama, Durant, the Spurs, and the notebooks' `game_id` `0042500405`). The results were then compared against each other value by value. Together these 145 calls are the whole library: 139 `stats` endpoints, 4 `live` and 2 `static`.

Verdict	Calls	What it means
Use	**80**	63 core calls, plus 17 optional ones that are useful but niche
Drop	**65**	28 return no data · 30 duplicate another call · 7 aren't analysis data

Start here

These calls cover most analyses. Everything else in this document is detail.

I want…	Call
Player / team IDs	`stats.static.players`, `stats.static.teams` (offline, no request)
A player table (bio, draft, college)	`playerindex(historical_nullable=1)`
Every game result in a season	`leaguegamelog` (one call, every team)
Every player's game log for a season	`playergamelogs` (one call, every player)
Season stats for all players	`leaguedashplayerstats`
Season stats for all teams	`leaguedashteamstats`
Tracking: speed, drives, passing, rebounding…	`leaguedashptstats`
Every shot with x/y	`shotchartdetail`
Every event in a game	`playbyplayv3`, or live `playbyplay` (richer)
A full box score for one game	`boxscoretraditionalv3` + the other `…v3` box scores
Lineups and on/off	`leaguedashlineups`, `teamplayeronoffsummary`
Standings	`leaguestandingsv3`
A player's career	`playercareerstats`
The schedule	`scheduleleaguev2`
Play types	`synergyplaytypes`

Tip: `leaguedashplayerstats` and `leaguedashteamstats` take `measure_type_detailed_defense` (Base, Advanced, Misc, Scoring, Usage, Four Factors, Defense) and `per_mode_detailed` (Totals, PerGame, Per36, …). One endpoint therefore gives you seven different stat tables.

Usable calls by topic

All calls are in `nba_api.stats.endpoints` unless noted.

Players & teams

Call	What you get
`stats.static.players`	Player name ↔ ID (5,198 players, offline)
`stats.static.teams`	Team abbreviation ↔ ID (offline)
`playerindex`	One row per player: position, height, weight, college, draft, years active
`commonplayerinfo`	One player's bio, including birthdate and seasons of experience
`commonteamroster`	Team roster for a season, plus the coaching staff
`playerawards`	Every award a player has won

Games, schedules & game logs

Call	What you get
`leaguegamelog`	Every game of a season, one row per team (`'T'`) or per player (`'P'`)
`playergamelogs`	Player game logs with 72 columns; all season types
`teamgamelogs`	Team game logs with 57 columns; all season types
`leaguegamefinder`	Search games across seasons with filters
`scheduleleaguev2`	The full season schedule (1,400 games)

Optional:

- `scoreboardv3` gives one date's games.
- `scoreboardv2` is worth keeping **only** for its conference standings on any date.
- `playergamestreakfinder` finds streaks such as 3+ games with 20+ points.

Box score for one game

All of these take `game_id`. They return one row per player, plus team totals.

Call	What you get
`boxscoretraditionalv3`	Counting stats, with starters vs bench
`boxscoreadvancedv3`	Ratings, pace, possessions, TS%, usage, PIE
`boxscorefourfactorsv3`	eFG%, FTA rate, TOV%, OREB% (own and opponent)
`boxscoremiscv3`	Points off turnovers, second-chance, fast-break and paint points
`boxscorescoringv3`	Shot mix and assisted/unassisted shares
`boxscoreusagev3`	Each player's share of team stats
`boxscoreplayertrackv3`	Speed, distance, touches, passes, contested shots
`boxscorehustlev2`	Deflections, charges, screen assists, box-outs
`boxscoredefensivev2`	Matchup defence per defender
`boxscorematchupsv3`	Who guarded whom, for how long, and the result
`boxscoresummaryv3`	Officials, arena, attendance, line score, last 5 meetings

`boxscorehustlev2` and `boxscoredefensivev2` are the current versions; there is no V3 for either.

Play-by-play & shots

Call	What you get
`playbyplayv3`	One row per event (24 columns)
`live.nba.endpoints.playbyplay`	Richer events (56 columns): x/y, possession, linked assist/steal/block. Works for past games too.
`shotchartdetail`	Every shot with location and zone. Pass `context_measure_simple='FGA'` to include misses.

Optional: `shotchartlineupdetail` returns the shots taken while one 5-man lineup was on the floor.

Season stats: every player

Call	What you get
`leaguedashplayerstats`	**The main season table** (see the tip above)
`leaguedashplayerbiostats`	Age, height, weight and draft for everyone who played (582 players)
`leaguedashplayerclutch`	Stats in clutch time
`leaguedashplayershotlocations`	FG by court zone
`leaguedashplayerptshot`	Shot tracking by closest defender, dribbles, touch time and shot clock
`leaguedashptdefend`	Opponent FG% when this player defends
`leaguehustlestatsplayer`	Season hustle stats
`playerestimatedmetrics`	Estimated ratings, pace and rebound/assist rates

Season stats: every team

Call	What you get
`leaguedashteamstats`	**The main team table**
`leaguedashptstats`	**Tracking.** `pt_measure_type` = SpeedDistance, Rebounding, Passing, Drives, CatchShoot, PullUpShot, Defense, Possessions, … Set `player_or_team='Player'` for player rows.
`leaguedashteamptshot`	Team shot tracking
`leaguedashoppptshot`	Opponent shot tracking
`leaguedashptteamdefend`	Opponent FG% when defended
`leaguedashteamshotlocations`	Team FG by zone
`leaguedashteamclutch`	Team clutch stats
`leaguehustlestatsteam`	Team hustle stats
`teamestimatedmetrics`	Team estimated ratings and pace

Splits (player & team dashboards)

Each dashboard returns an "Overall" row that's identical to `leaguedashplayerstats` / `leaguedashteamstats`. Only the split tables are new.

Call	Splits
`playerdashboardbygeneralsplits`	Home/away, wins/losses, month, before/after the All-Star break, starter/bench, rest days
`playerdashboardbygamesplits`	By half, quarter and score margin
`playerdashboardbyshootingsplits`	By distance, area, shot type, and **who assisted**
`playerdashptpass`	Passes to and from each teammate (a pass network)
`teamdashboardbygeneralsplits`	The team version of general splits. Don't use `plus_minus='Y'`: it returns wrong values.
`teamdashboardbyshootingsplits`	The team version of shooting splits

Optional:

- `playerdashboardbyclutch` gives 10 different clutch definitions.
- `playerdashboardbylastngames` covers the last 5/10/15/20 games.
- `playerdashboardbyteamperformance` splits by team score differential.
- `playerdashboardbyyearoveryear` is per-season and overlaps `playercareerstats`.

Lineups, on/off & matchups

Call	What you get
`leaguedashlineups`	Every lineup's stats; filter by team with `team_id_nullable`
`teamplayeronoffsummary`	Team net, offensive and defensive rating with each player on vs off
`teamplayeronoffdetails`	The full stat line with each player on vs off
`leagueseasonmatchups`	Season-long head-to-head matchup stats
`synergyplaytypes`	Play types (Isolation, Transition, P&R, …): points per possession, frequency, percentile

Optional:

- `leaguelineupviz` gives lineup ratings with a minutes cut-off.
- `matchupsrollup` shows who a defender guarded, by position.
- `playervsplayer` gives stats with a specific opponent on vs off the floor.

Career, history & franchises

Call	What you get
`playercareerstats`	Season and career totals (regular season, playoffs, All-Star, college), plus highs
`teamyearbyyearstats`	A franchise's every season: record and playoff result
`franchisehistory`	Totals for every franchise, including defunct ones
`franchiseplayers`	Every player's totals with one franchise
`teamdetails`	Arena, owner, titles, Hall of Famers, retired numbers

Optional: `alltimeleadersgrids` lists all-time career leaders by stat.

Standings & playoffs

Call	What you get
`leaguestandingsv3`	Standings with seeding, clinch flags, home/road and splits
`iststandings`	NBA Cup group standings
`commonplayoffseries`	Playoff games with series ID and game number

Draft & combine

Call	What you get
`drafthistory`	Every pick: round, number, team, school
`draftcombinestats`	Combine measurements, athletic drills and shooting in one table

Optional: `draftcombinespotshooting` and `draftcombinenonstationaryshooting` give made/attempted shots per spot. Both are sparse: only 43 of 81 players have data.

New tracking metrics (optional)

- `gravityleaders` gives each player's gravity score: how much defensive attention the player draws.
- `dunkscoreleaders` has one row per dunk, with vertical, hang time and style scores.

Live

- `live.nba.endpoints.boxscore` (optional) is for games **in progress**. For finished games it repeats `boxscoretraditionalv3`.

Calls to drop

Return no data (28)

The server sends nothing back. These 11 are deprecated in nba_api. They return either `{}` or the NBA.com homepage instead of data.

Call	Use instead
`boxscoreadvancedv2`, `boxscorefourfactorsv2`, `boxscoremiscv2`, `boxscorescoringv2`, `boxscoreusagev2`	The matching `…v3`
`playbyplay`, `playbyplayv2`	`playbyplayv3`
`leagueleaders`, `assistleaders`	`leaguedashplayerstats`
`playoffpicture`	`leaguestandingsv3`
`assisttracker`	—

Server error (HTTP 500) for every input tried. There are 15 of these. Many hang for about 30 seconds before failing. nba_api doesn't mark them deprecated, so they might come back. Re-test them later.

Call	Use instead
`playerdashptreb`, `teamdashptreb`, `teamdashptpass`	`leaguedashptstats` (Rebounding / Passing)
`playerdashptshots`, `teamdashptshots`	`leaguedashplayerptshot`, `leaguedashteamptshot`
`gamerotation`	Substitution events in live `playbyplay`
`teamgamestreakfinder`	Compute from `teamgamelogs`
`defensehub`	`leaguedashptdefend`
`teamandplayersvsplayers`	`playervsplayer`
`draftboard`	`drafthistory`
`playercareerbycollege`	`playerindex` (filter on `COLLEGE`)
`teamhistoricalleaders`	`franchiseplayers`
`winprobabilitypbp`, `glalumboxscoresimilarityscore`, `videodetails`	—

Return 0 rows. Both were tried on two seasons.

Call	Use instead
`playerdashptshotdefend`	`leaguedashptdefend`

Call	Use instead
`playerfantasyprofilebargraph`	`leaguedashplayerstats`

Duplicates (30)

These still work, but another call returns the same data. Rows marked *(verified)* were compared value by value.

Drop	Use instead	Why
`boxscoretraditionalv2`	`boxscoretraditionalv3`	Identical *(verified)*; deprecated
`boxscoresummaryv2`	`boxscoresummaryv3`	Mostly empty for games after 2025-04-10
`hustlestatsboxscore`	`boxscorehustlev2`	Identical *(verified)*
`playergamelog`	`playergamelogs`	Subset *(verified)*; regular season only
`teamgamelog`	`teamgamelogs`	Subset; its `TOV` leaves out team turnovers
`cumestatsplayergames`, `cumestatsteamgames`	game logs	Just a list of game IDs *(verified)*
`cumestatsplayer`, `cumestatsteam`	game logs	Sums over chosen games; aggregate the logs yourself
`playerprofilev2`	`playercareerstats`	Same career tables *(verified)*
`commonallplayers`	`playerindex`	Same players; fewer columns
`commonteamyears`	`franchisehistory`	A team list with first/last year
`teaminfocommon`	`leaguestandingsv3`	One team's record and 4 ranks
`franchiseleaders`	`franchiseplayers`	Its top row *(verified)*
`teamplayerdashboard`	`leaguedashplayerstats(team_id_nullable=…)`	Identical *(verified)*
`teamdashlineups`	`leaguedashlineups`	Part of the league call's output *(verified)*
`leagueplayerondetails`	`teamplayeronoffdetails`	Identical *(verified)*
`leaguestandings`	`leaguestandingsv3`	V3 has 13 more columns
`scheduleleaguev2int`	`scheduleleaguev2`	Same games; adds international broadcasters
`playernextngames`	`scheduleleaguev2`	The next few games on the schedule
`shotchartleaguewide`	`shotchartdetail`	Its league-average table
`homepageleaders`, `homepagev2`, `leaderstiles`	`leaguedashplayerstats`	Home-page top-5 widgets
`playercompare`	`leaguedashplayerstats`	Two players' rows side by side
`draftcombineplayeranthro`, `draftcombinedrillresults`	`draftcombinestats`	Identical subsets *(verified)*
`fantasywidget`, `infographicfanduelplayer`	`leaguedashplayerstats` / box scores	Fantasy points are a formula over box-score stats
live `scoreboard`	`scoreboardv3`	Today's games only

Not analysis data (7)

- `videodetailsasset`, `videoevents`, `videoeventsasset` and `videostatus` return video links and availability flags, not stats.
- live `odds` has today's betting odds only, with no history.
- `teamvsplayer` ignores `team_id` and returns all 30 teams, and its on/off tables are empty.
- `playercareerbycollegerollup` puts colleges into an NCAA-style bracket; a novelty.

Watch out for

1. **The notebooks' `game_id` is a playoff game.** `leaguegamefinder` returns every season type by default, so "the last game of the season" is Finals Game 5. Pass `season_type_nullable='Regular Season'` if you want the regular season.
2. **Dates come in four formats:** `Apr 10, 2026`, `APR 12, 2026`, `2026-06-13T00:00:00` and `2026-06-13`. Parse them before joining tables.
3. **`MIN` means different things:** `"39:10"` in box scores, 240 player-minutes in `teamgamelog`, and 48.0 game minutes in `teamgamelogs`.
4. **V2 box scores leave stats blank (NaN) for players who didn't play; V3 uses 0.** Fill with 0 before summing.
5. **`playerindex(season=…)` returns only 241 players.** For everyone who played that season, use `leaguedashplayerbiostats` (582).

6. `leaguegamefinder(game_id_nullable=…)` **is ignored by the API.** Filter by game ID in pandas instead.
7. **Deprecated calls will be removed.** nba_api already warns on 13 endpoints. All of them are in the drop lists except `scoreboardv2`.

Notebook fixes

- `player-comparisons-and-matchups.ipynb` uses `playergamelog` without importing it, so it raises `NameError`.
- `shot-charts.ipynb` uses `leaguedashlineups` without importing it, so it raises `NameError`.
- The same setup cell is pasted into 24 notebooks. Load it once from `example-inputs/inputs.ipynb` instead.
- None of the notebooks have saved outputs, so these failures weren't visible.

Appendix: every call A–Z

Call	Verdict	Use instead
alltimeleadersgrids	Optional	
assistleaders	Dead	leaguedashplayerstats
assisttracker	Dead	
live boxscore	Optional	
boxscoreadvancedv2	Dead	boxscoreadvancedv3
boxscoreadvancedv3	Core	
boxscoredefensivev2	Core	
boxscorefourfactorsv2	Dead	boxscorefourfactorsv3
boxscorefourfactorsv3	Core	
boxscorehustlev2	Core	
boxscorematchupsv3	Core	
boxscoremiscv2	Dead	boxscoremiscv3
boxscoremiscv3	Core	
boxscoreplayertrackv3	Core	
boxscorescoringv2	Dead	boxscorescoringv3
boxscorescoringv3	Core	
boxscoresummaryv2	Duplicate	boxscoresummaryv3
boxscoresummaryv3	Core	
boxscoretraditionalv2	Duplicate	boxscoretraditionalv3
boxscoretraditionalv3	Core	
boxscoreusagev2	Dead	boxscoreusagev3
boxscoreusagev3	Core	
commonallplayers	Duplicate	playerindex
commonplayerinfo	Core	
commonplayoffseries	Core	
commonteamroster	Core	
commonteamyears	Duplicate	franchisehistory
cumestatsplayer	Duplicate	playergamelogs
cumestatsplayergames	Duplicate	playergamelogs
cumestatsteam	Duplicate	teamgamelogs
cumestatsteamgames	Duplicate	teamgamelogs
defensehub	Dead	leaguedashptdefend
draftboard	Dead	drafthistory
draftcombinedrillresults	Duplicate	draftcombinestats
draftcombinenonstationaryshooting	Optional	
draftcombineplayeranthro	Duplicate	draftcombinestats
draftcombinespotshooting	Optional	
draftcombinestats	Core	
drafthistory	Core	
dunkscoreleaders	Optional	
fantasywidget	Duplicate	leaguedashplayerstats
franchisehistory	Core	
franchiseleaders	Duplicate	franchiseplayers
franchiseplayers	Core	
gamerotation	Dead	live playbyplay
glalumboxscoresimilarityscore	Dead	
gravityleaders	Optional	
homepageleaders	Duplicate	leaguedashplayerstats
homepagev2	Duplicate	leaguedashplayerstats
hustlestatsboxscore	Duplicate	boxscorehustlev2
infographicfanduelplayer	Duplicate	box scores
iststandings	Core	
leaderstiles	Duplicate	leaguedashplayerstats
leaguedashlineups	Core	

Call	Verdict	Use instead
leaguedashoppptshot	Core	
leaguedashplayerbiostats	Core	
leaguedashplayerclutch	Core	
leaguedashplayerptshot	Core	
leaguedashplayershotlocations	Core	
leaguedashplayerstats	Core	
leaguedashptdefend	Core	
leaguedashptstats	Core	
leaguedashptteamdefend	Core	
leaguedashteamclutch	Core	
leaguedashteamptshot	Core	
leaguedashteamshotlocations	Core	
leaguedashteamstats	Core	
leaguegamefinder	Core	
leaguegamelog	Core	
leaguehustlestatsplayer	Core	
leaguehustlestatsteam	Core	
leagueleaders	Dead	leaguedashplayerstats
leaguelineupviz	Optional	
leagueplayerondetails	Duplicate	teamplayeronoffdetails
leagueseasonmatchups	Core	
leaguestandings	Duplicate	leaguestandingsv3
leaguestandingsv3	Core	
matchupsrollup	Optional	
live odds	Not data	
live playbyplay	Core	
playbyplay	Dead	playbyplayv3
playbyplayv2	Dead	playbyplayv3
playbyplayv3	Core	
playerawards	Core	
playercareerbycollege	Dead	playerindex
playercareerbycollegerollup	Not data	
playercareerstats	Core	
playercompare	Duplicate	leaguedashplayerstats
playerdashboardbyclutch	Optional	
playerdashboardbygamesplits	Core	
playerdashboardbygeneralsplits	Core	
playerdashboardbylastngames	Optional	
playerdashboardbyshootingsplits	Core	
playerdashboardbyteamperformance	Optional	
playerdashboardbyyearoveryear	Optional	
playerdashptpass	Core	
playerdashptreb	Dead	leaguedashptstats
playerdashptshotdefend	Dead	leaguedashptdefend
playerdashptshots	Dead	leaguedashplayerptshot
playerestimatedmetrics	Core	
playerfantasyprofilebargraph	Dead	leaguedashplayerstats
playergamelog	Duplicate	playergamelogs
playergamelogs	Core	
playergamestreakfinder	Optional	
playerindex	Core	
playernextngames	Duplicate	scheduleleaguev2
playerprofilev2	Duplicate	playercareerstats
static.players	Core	

Call	Verdict	Use instead
playervsplayer	Optional	
playoffpicture	Dead	leaguestandingsv3
scheduleleaguev2	Core	
scheduleleaguev2int	Duplicate	scheduleleaguev2
live scoreboard	Duplicate	scoreboardv3
scoreboardv2	Optional	
scoreboardv3	Optional	
shotchartdetail	Core	
shotchartleaguewide	Duplicate	shotchartdetail
shotchartlineupdetail	Optional	
synergyplaytypes	Core	
teamandplayersvsplayers	Dead	playervsplayer
teamdashboardbygeneralsplits	Core	
teamdashboardbyshootingsplits	Core	
teamdashlineups	Duplicate	leaguedashlineups
teamdashptpass	Dead	leaguedashptstats
teamdashptreb	Dead	leaguedashptstats
teamdashptshots	Dead	leaguedashteamptshot
teamdetails	Core	

Call	Verdict	Use instead
teamestimatedmetrics	Core	
teamgamelog	Duplicate	teamgamelogs
teamgamelogs	Core	
teamgamestreakfinder	Dead	teamgamelogs
teamhistoricalleaders	Dead	franchiseplayers
teaminfocommon	Duplicate	leaguestandingsv3
teamplayerdashboard	Duplicate	leaguedashplayerstats
teamplayeronoffdetails	Core	
teamplayeronoffsummary	Core	
static.teams	Core	
teamvsplayer	Not data	
teamyearbyyearstats	Core	
videodetails	Dead	
videodetailsasset	Not data	
videoevents	Not data	
videoeventsasset	Not data	
videostatus	Not data	
winprobabilitypbp	Dead	